

PsyVEC: A Self-Evolving Multi-Agent Framework for Depression Assessment through Verified Experience Consolidation

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Abstract

Large language models have shown growing potential for automated depression assessment, particularly when multiple specialized agents collaborate to acquire symptom evidence, evaluate its sufficiency, score standardized items, and maintain a coherent assessment state. However, most existing multi-agent systems remain fixed after development and cannot systematically improve from previous cases. Directly reusing self-generated experience is also unreliable because past interactions may contain noisy, case-specific, or misleading feedback. We propose PsyVEC (Psychometric Verified Experience Consolidation), a self-evolving multi-agent framework for depression assessment. PsyVEC identifies informative decisions from prior cases, verifies whether the resulting experience provides consistent benefit, and consolidates reliable knowledge for future assessments. By associating experience with the responsible agent roles, the framework supports coordinated improvement while reducing the propagation of unreliable feedback across the system. We evaluate PsyVEC with two Qwen models against prompting-based and domain-specific baselines. The evaluation covers depression-assessment accuracy, evidence acquisition, efficiency, generalization, component contributions, reliability of accumulated experience, computational trade-offs, robustness, and human-perceived interaction quality. Overall, PsyVEC provides a controlled and auditable approach for developing multi-agent depression-assessment systems that improve through accumulated experience. It is intended for offline research and does not replace professional clinical diagnosis or autonomously learn from live patients.

Keywords: self-evolving agents, multi-agent systems, large language models, mental health assessment, experience memory, direct preference optimization

1. Introduction

Mental disorders remain a major and growing public-health burden, affecting nearly one in eight people worldwide and accounting for substantial disability across countries and age groups (World Health Organization, 2022; GBD 2019 Mental Disorders Collaborators, 2022). Timely assessment is difficult to scale because useful evidence is often embedded in free-text disclosures and must be interpreted across multiple symptoms, time frames, and functional impacts. Recent mental-health LLMs can analyze large collections of online text, produce interpretable mental-health inferences, and map unstructured interviews to structured psychiatric questionnaires (Xu et al., 2024; Yang et al., 2024; Rosenman et al., 2024). These capabilities make LLMs promising tools for scalable *screening support*; they do not make them autonomous diagnostic systems.

For depression assessment, however, screening is not only a problem of interpreting a completed record. Standardized instruments such as the Patient Health Questionnaire (PHQ-8) require evidence about several symptom dimensions over a defined period (Kroenke et al., 2009). A participant may

mention poor sleep while leaving its frequency, persistence, or impact unclear. If the system never asks the relevant follow-up, even an accurate downstream analyzer cannot recover the missing evidence. The central challenge is therefore to acquire the right evidence as the conversation unfolds, while keeping the interaction focused, efficient, and safe.

Interactive LLM systems begin to address this challenge. AgentMental coordinates questioning, evidence-adequacy evaluation, scoring, and state updates for scale-based assessment (Hu et al., 2026); MAGI organizes psychiatric interviewing through specialized agents aligned with a diagnostic protocol (Bi et al., 2025); and WiseMind combines complementary reasoning agents with psychiatric knowledge (Wu et al., 2026). Together, these systems show that role-specialized agents can ask targeted follow-ups and maintain interpretable interview state. Yet their competence is largely fixed across cases: they adapt within the current interview but do not establish which decisions from a past interview should reliably improve a future one.

Enabling cross-case improvement is appealing when labeled clinical data are limited, but naively storing successful trajectories is unsafe. A participant may provide unusually explicit answers regardless of the question asked, so a fluent post-hoc reflection can mistake coincidence for a reusable interviewing rule. The same issue affects parameter updates: a terminal as-

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assessment error does not identify whether questioning, evidence evaluation, scoring, or state revision caused it. General language agents can distill feedback into experience memory and retrieve it in later tasks (Shinn et al., 2023; Zhao et al., 2024; Ouyang et al., 2025), while preference optimization provides a practical route from pairwise choices to model updates (Rafailov et al., 2023). For mental-health interviewing, though, both forms of learning require stronger evidence: a local action preference and a transferable natural-language lesson make different claims and should not be accepted on the basis of the same unverified trajectory.

We introduce **PsyVEC**, a self-evolving framework for role-specialized depression-assessment agents. PsyVEC first identifies informative interview states, then compares alternative actions by replaying them from the same pre-action state with the same frozen simulated participant profile. This controlled simulator-level intervention isolates the agent decision from participant variation; it is not a causal claim about real patients. A role-conditioned utility evaluates task improvement, critical-evidence gain, interaction cost, redundancy, and safety. Only contrasts that meet explicit consistency and utility-margin criteria can become role-specific preference data.

PsyVEC separately tests whether a valid local contrast yields a reusable procedural lesson. A distilled lesson is inserted into matched held-out states under fixed policy, simulator, and decoding conditions, and enters cross-case memory only when paired replay shows positive transfer without a safety regression. This separation yields two complementary learning paths: verified lessons can support retrieval at inference time, whereas verified local preferences can update only the responsible role through LoRA-DPO (Hu et al., 2022). The framework further uses responsibility-aware routing, participant-disjoint regression tests, and memory-ON/OFF evaluation to make each improvement auditable.

The main contributions are as follows:

1. **Controlled psychometric experience generation.** We introduce role-aware hard-state mining and matched same-state branching under a frozen participant profile, producing decision contrasts in which participant variation is controlled rather than treating successful trajectories as trustworthy supervision.
2. **Dual-gate, dual-path consolidation.** We distinguish state-local preference validity from cross-state lesson transfer, using locally verified pairs for role-specific LoRA-DPO and admitting procedural lessons to memory only after source-excluded paired validation.
3. **Localized and auditable self-evolution.** We combine responsibility-aware routing, participant-disjoint regression gates, memory-ON/OFF evaluation, and lesson-consistent action analysis to identify retrieval gains, memory-independent policy gains, and cross-role regressions.
4. **A claim-matched empirical protocol.** We formulate a seven-study evaluation across Qwen3.5 9B and 27B backbones, separating interactive baselines from passive full-transcript references and testing verification, hard-state al-

location, role localization, internalization, computational trade-offs, robustness, and failure-to-success transitions without fabricating cross-protocol comparisons.

PsyVEC is designed as an offline research framework with simulated participants. It is intended to study how evidence-verified experience can improve assessment support, not to replace professional diagnosis or autonomously learn from live patients.

2. Related work

To be completed in the next writing stage.

3. Proposed method: PsyVEC

PsyVEC, short for *Psychometric Verified Experience Consolidation*, is an independent multi-agent framework for interactive mental-health assessment and closed-loop offline evolution. Its governing principle is that an interaction experience may affect reusable memory or model parameters only after controlled replay supports the utility of the corresponding decision. An interview first produces role-tagged trajectories. A hard-state miner then identifies decisions worth additional compute, and same-state branching evaluates alternative actions against a frozen participant profile. This common experience factory yields a high-confidence matched contrast

$$C = (s, r, a^+, a^-, \Delta U), \quad \Delta U = U_r(s, a^+) - U_r(s, a^-). \quad (1)$$

The contrast enters two consolidation paths (Fig. 1). The fast path distills a procedural lesson and subjects it to cross-state transfer verification before retrieval. The slow path treats the same locally verified actions as a role-specific preference pair for LoRA-DPO. Crucially, the paths are not coupled: a pair may provide credible local supervision even when its linguistic abstraction fails to transfer. No raw trajectory is directly consolidated.

3.1. Interactive multi-agent formulation and knowledge segregation

3.1.1. Role-specialized assessment

Let participant p be assessed over the topics $\mathcal{T} = \{T_1, \dots, T_m\}$ of a questionnaire; $m = 8$ for PHQ-8. We design four functional decision roles,

$$\mathcal{A} = \{A_i, A_e, A_s, A_u\}, \quad (2)$$

corresponding to interviewing, evidence-sufficiency evaluation, item scoring, and belief-state updating. These are functional policies rather than four copies of the backbone. At turn j for topic T_i , the shared observable interaction state is

$$s_{i,j} = (T_i, H_{i,j}, M_{i,j}^{\text{case}}, E_{i,j}^{\text{known}}, E_{i,j}^{\text{miss}}), \quad (3)$$

where $H_{i,j}$ is the dialogue history, $M_{i,j}^{\text{case}}$ is the within-participant memory, and $E_{i,j}^{\text{known}}$ and $E_{i,j}^{\text{miss}}$ are acquired and missing evidence slots. For the PHQ setting, slots include symptom presence, frequency, persistence, context, and functional impact. The

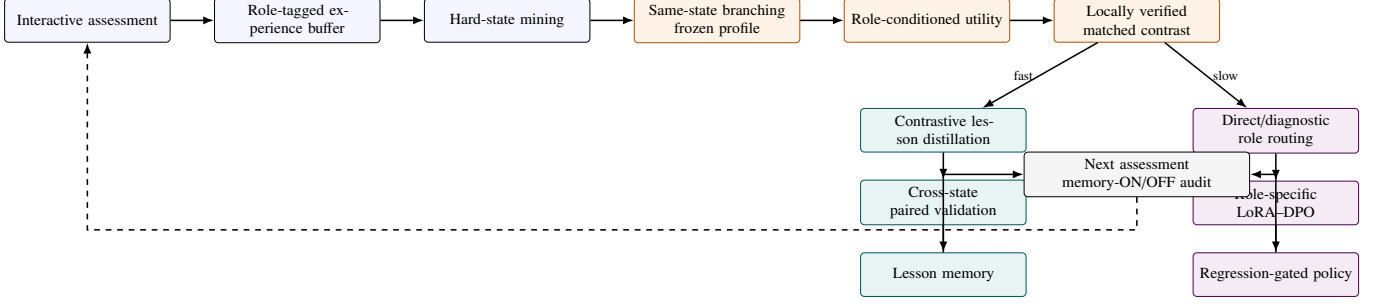


Figure 1: PsyVEC converts controlled interaction contrasts into two verified consolidation paths. The fast path promotes transferable lessons into external memory after cross-state paired validation; the slow path converts locally verified action preferences into role-specific LoRA-DPO updates. The two paths share an evidence source but impose verification appropriate to different generalization claims.

assessment policy never observes the ground-truth PHQ item score.

An action depends on role: A_i generates the next question, A_e selects continue/stop (or a sufficiency judgment), A_s assigns an item score, and A_u updates a prior summary or score. A frozen backbone is shared across roles, while role prompts and low-rank adapters separate behavior:

$$a^r \sim \pi_r(a \mid s; \phi_r), \quad r \in \{i, e, s, u\}, \quad (4)$$

where only adapter parameters ϕ_r are trainable in the slow path. Adapter switching therefore implements specialization without deploying four full LLMs.

3.1.2. Three-timescale knowledge segregation

PsyVEC separates information by lifetime and purpose (Table 1). *Case memory* stores the current participant’s evidence, topic summaries, scores, and history, and is destroyed or isolated after the case. *Lesson memory* stores only transfer-validated procedural rules and is retrieved across participants. *Policy adapters* contain behavior that has been consolidated parametrically. Calling all three “memory” would obscure their different access and validation requirements.

The experience buffer records tuples $(s, r, a, o, s', U, \rho)$, where o is the simulator response or role output, s' the resulting state, U a decomposed offline utility, and ρ provenance including participant, policy, seed, and simulator version. Raw identifiers and verbatim participant data are not made available to future cases. This segregation blocks a simple leakage channel and permits every promoted lesson and preference pair to be traced to the replay evidence that created it.

3.2. Controlled experience exploration

Most interaction turns proceed under the accepted policy without branching. PsyVEC spends extra simulator calls only where the expected learning value is high. Controlled exploration has three steps: mine a hard role decision, replay diverse actions under matched conditions, and retain only locally verified contrasts.

3.2.1. Role-aware hard-state mining

For a state in which role r must act, the miner combines three normalized and auditable signals:

$$h_r(s) = \alpha m(s) + \beta u_r(s) + \gamma c_r(s), \quad \alpha + \beta + \gamma = 1. \quad (5)$$

Here, $m(s)$ is the weighted fraction of critical evidence still missing; $u_r(s)$ is role-specific decision uncertainty; and $c_r(s)$ is coordination risk. Examples of u_r include normalized scorer entropy and proximity to the evaluator’s stop threshold. Coordination risk covers evaluator–scorer disagreement, semantic duplication of a proposed question, and inconsistency between new evidence and the stored state. The system branches when $h_r(s) \geq \tau_h$. All weights and τ_h are fixed on the participant-disjoint development set. A rule-based miner is preferred to an additional learned predictor because its decisions are inspectable and the evolution corpus is small. A matched-compute comparison with random-state branching is required to establish that any gain is not merely due to additional simulator calls.

3.2.2. Controlled same-state branching

At a selected state, PsyVEC samples K plausible actions and replays them from identical conditions:

$$\mathcal{R}_r(s, z_p) = \{(a_k, o_k, s'_k)\}_{k=1}^K, \quad s_k = s, \quad z_{p,k} = z_p \quad \forall k, \quad (6)$$

where z_p is a structured frozen participant profile created once before branching. In the PHQ setting it contains transcript-supported facts for each item—presence, frequency, context, impact, and the hidden item score. Policies cannot access the hidden fields; only the participant simulator and offline evaluator can. If a fact is unsupported by the transcript, the simulator expresses uncertainty rather than inventing a symptom. The default $K = 4$ candidates comprise a low-temperature action from the current policy, an exploratory sample, a diversity-prompted sample, and a lesson-guided action when a compatible lesson exists. This composition is an implementation choice, not the methodological claim; the essential property is matched evaluation with fixed s and z_p .

Deterministic simulation is used for the primary comparison. For stochastic sensitivity analysis, competing branches share common random seeds. Each branch is rejected if its response

Table 1: Knowledge segregation in PsyVEC. The experience buffer is an audit/training store, not an inference-time knowledge layer.

Component	Lifetime	Content and purpose	Inference access
Case memory M^{case}	One participant	Evidence slots, dialogue history, topic summaries, and current scores	Direct, current case only
Lesson memory M^{lesson}	Cross-case	Transfer-validated conditional procedural lessons	Retrieved by role and scope
Policy adapters Φ	Persistent	Internalized role behavior encoded in LoRA parameters	Parametric
Experience buffer $\mathcal{B}$	Evolution cycle	Role-tagged state transitions, utilities, provenance, and simulator diagnostics	Never inserted into prompts

contradicts the frozen profile or if the simulator introduces unsupported clinical facts. Reporting profile-consistency rate, contradiction rate, and the fraction of invalidated branches makes the reliability of this controlled simulator-level intervention visible.

3.2.3. Role-conditioned utility and local contrast verification

A single question-centric reward is unsuitable for all four roles. PsyVEC instead uses a shared utility skeleton with role-specific task gains:

$$U_r(s, a) = \lambda_r G_r^{\text{task}}(s, a) + \lambda_e G_r^{\text{evid}}(s, a) - \lambda_c C_r(s, a), \quad \text{Safe}(s, a) = 1. \quad (7)$$

The safety constraint is a hard admissibility gate, not a small additive term. Table 2 specifies the role semantics. For a question, the utility can be instantiated as

$$U_i = \lambda_s G_{\text{score}} + \lambda_e G_{\text{evid}} + \lambda_r R_{\text{rel}} - \lambda_d C_{\text{dup}} - \lambda_t C_{\text{turn}}, \quad (8)$$

where $G_{\text{evid}} = \text{Cov}_{\text{crit}}(s') - \text{Cov}_{\text{crit}}(s)$ and critical coverage is a weighted fraction of required evidence slots. Information acquisition as a question-utility signal is consistent with recent proactive medical LLM work (Ding et al., 2026), but PsyVEC uses a lightweight scale-grounded definition rather than a learned holistic judge. With item target $y_i \in \{0, 1, 2, 3\}$ and estimates before/after the action, scoring gain is

$$G_{\text{score}} = \text{clip}\left(\frac{|\hat{y}_i^- - y_i| - |\hat{y}_i^+ - y_i|}{3}, -1, 1\right). \quad (9)$$

The label is hidden environment supervision: it is never exposed in s , candidate generation, or lesson text. An evidence-only utility ablation tests dependence on this supervision.

Among safe valid branches, $a^+ = \arg \max_k U_r(s, a_k)$. The negative a^- is a lower-utility but plausible action, not a malformed output that the policy could reject trivially. A pair passes *local contrast verification* when

$$\Delta U \geq \delta_{\text{pair}} \wedge \text{Consistent} \wedge \text{NonDuplicate} \wedge \text{Attributed}(r). \quad (10)$$

This gate establishes a state-local preference. Filtering ambiguous pairs is important because preference optimization is sensitive to noisy ordering (Ray Chowdhury et al., 2024); sampling alternatives from the accepted policy also keeps the pair near the policy distribution (Tajwar et al., 2024). The gate does *not* establish that a natural-language rule inferred from the pair transfers to other participants.

3.3. Fast evolution: verified lesson consolidation

3.3.1. Contrastive lesson distillation

The fast path converts a matched contrast into an interpretable candidate lesson. Rather than reflecting on one failed trajectory, the distiller receives (s, a^+, o^+, U^+) and (s, a^-, o^-, U^-) under their common starting condition. It emits

$$\ell = (r, c, a^{\text{do}}, a^{\text{avoid}}, e, \sigma, q), \quad (11)$$

where r is the target role, c a trigger, a^{do} the recommended behavior, a^{avoid} the behavior to avoid, e the target evidence or criterion, σ the scope and exceptions, and q the validation confidence. Provenance and lifecycle state remain metadata rather than expanding the mathematical object. A canonical realization is: IF trigger, DO action, AVOID alternative, BECAUSE criterion, EXCEPT WHEN scope condition. The contribution is controlled contrastive induction, not the general idea of learning from successes and failures (Ouyang et al., 2025).

3.3.2. Cross-state paired validation

Distillation produces a candidate, not production memory. For lesson ℓ , the validator retrieves historical evolve states

$$\mathcal{S}_\ell = \{s_1, \dots, s_{n_\ell}\} \quad (12)$$

whose role, trigger, evidence pattern, and topic/scope are compatible. The source participant is excluded. Each state is replayed from the same frozen profile in two conditions. The control uses the accepted policy with the current lesson memory M^{lesson} ; the intervention is identical except that candidate ℓ is made available. Thus, the comparison isolates the marginal contribution of ℓ rather than comparing no memory against augmented memory. With common randomness where necessary,

$$\Delta_j(\ell) = U_{\text{int}}(s_j; M^{\text{lesson}} \cup \{\ell\}) - U_{\text{ctrl}}(s_j; M^{\text{lesson}}). \quad (13)$$

Transfer verification promotes ℓ only if

$$n_\ell \geq m_{\min}, \quad \frac{1}{n_\ell} \sum_j \mathbb{I}[\Delta_j(\ell) > 0] \geq \tau_w, \quad \overline{\Delta U}_\ell \geq \epsilon, \quad (14)$$

and no replay introduces a safety regression. The thresholds are tuned only on development participants. Confidence intervals are reported as analysis rather than used as a brittle hard gate when a trigger has few matched states. This second verification level supports a stronger claim than Eq. (10): the induced rule repeatedly helps outside its source case.

Table 2: Role-conditioned utility signals. The table defines which decision is credited, preventing a final error from becoming undifferentiated supervision for all agents.

Role	Positive task/evidence signal	Principal cost or failure
Interviewer	Critical-evidence and item-score gain; criterion relevance	Duplicate question and unnecessary turn
Sufficiency evaluator	Correctly continue when critical evidence is missing; stop when sufficient	Premature stop or unnecessary continuation
Scorer	Reduction in normalized item-score error	Error despite sufficient evidence
Updater	Correction of a stale or inconsistent prior state	Unsupported or omitted revision

3.3.3. Retrieval and lightweight governance

Before role r acts, the retriever hard-filters lessons by role and compatible trigger/scope. It ranks the remainder using only semantic match and validated confidence,

$$R(\ell | s) = \text{sim}(c_\ell, s) q_\ell, \quad (15)$$

and inserts at most K_L lessons to bound prompt growth. Governance deliberately uses three auditable operations. Near-duplicate lessons are merged with their evidence; unresolved conflicting recommendations are quarantined rather than retrieved; and a lesson is retired when its recent paired benefit is non-positive or the evolved policy has internalized it so that retrieval provides no additional benefit. All changes preserve validation records and provenance.

3.4. Slow evolution: role-localized preference optimization

3.4.1. Preference construction and role routing

Every locally verified contrast directly yields

$$d = (s, r, a^+, a^-). \quad (16)$$

This is self-generated preference supervision without *additional human preference annotation*; it is not label-free learning because the offline utility may use questionnaire labels. Importantly, a preference pair is admitted after local verification even when the corresponding lesson fails transfer verification. This prevents the fast path’s cross-case claim from becoming an unnecessary bottleneck for the slow path.

Routing has two stages. If branching perturbs an explicit decision by role r , responsibility is direct and the pair enters $\mathcal{D}_r$. Attribution analysis is required only for ambiguous episode-level failures. A diagnostic map first associates missing critical evidence with A_i , stopping with insufficient evidence with A_e , scoring error despite sufficient evidence with A_s , and failure to revise stale state with A_u . If several roles remain plausible, counterfactual component replay replaces the candidate role’s decision while holding other role *parameters* fixed and recomputing downstream outputs:

$$C_r = U(\text{Replay}(s; \tilde{a}_r, \pi_{-r})) - U(\text{Replay}(s; a_r, \pi_{-r})). \quad (17)$$

This follows the intervention logic of counterfactual multi-agent credit assignment (Foerster et al., 2018) without claiming a causal effect on real participants. Samples with no unique role above the attribution margin are excluded from parameter updates.

3.4.2. Role-specific LoRA–DPO

For each role, the accumulated dataset is $\mathcal{D}_r = \{(s_n, a_n^+, a_n^-)\}_{n=1}^{N_r}$. Starting from the currently accepted policy $\pi_r^{(e)}$, PsyVEC optimizes only ϕ_r with the standard DPO objective (Rafailov et al., 2023):

$$\mathcal{L}_{\text{DPO}}^{(r)} = -\mathbb{E}_{\mathcal{D}_r} \log \sigma \left(\beta \left[\log \frac{\pi_r(a^+ | s)}{\pi_{\text{ref},r}(a^+ | s)} - \log \frac{\pi_r(a^- | s)}{\pi_{\text{ref},r}(a^- | s)} \right] \right). \quad (18)$$

At evolution round e , $\pi_{\text{ref},r} = \pi_r^{(e)}$; after a candidate passes the development gate, it becomes the accepted reference for round $e + 1$. This makes each update a local consolidation relative to the last accepted behavior rather than anchoring all rounds to the initial model. The shared backbone and adapters $\phi_{r'}$ for $r' \neq r$ remain frozen. If $N_r < N_{\min}$, that role is not updated in the current round. High-confidence coverage is preferred to forcing four updates from sparse, uneven data. Iterative self-generated preferences have precedent (Yuan et al., 2024); PsyVEC’s distinctive role is to control how the pairs are generated, verified, and localized.

3.4.3. Regression gating and internalization verification

A lower training loss does not authorize deployment of a candidate adapter. Let Q denote assessment quality, C interaction cost, and R redundant-question rate on participant-disjoint development data. The accepted system maximizes

$$J(\Pi) = Q(\Pi, D_{\text{dev}}) - \eta_c C(\Pi, D_{\text{dev}}) - \eta_r R(\Pi, D_{\text{dev}}), \quad (19)$$

subject to no safety regression and predefined non-inferiority tolerances for roles not updated. A candidate checkpoint replaces $\Pi^{(e)}$ only if the gate passes; otherwise, it is rejected with its training provenance retained for audit.

Finally, PsyVEC distinguishes retrieved from internalized improvement. For held-out participants, the initial and evolved policies are evaluated with lesson memory both enabled and disabled. Parametric gain is

$$G_{\text{param}} = Q(\Pi^{(E)}, M = \emptyset) - Q(\Pi^{(0)}, M = \emptyset). \quad (20)$$

Memory-ON gain tests the combined system, whereas Eq. (20) tests what persists without retrieval. On held-out states satisfying a validated lesson trigger, the internalized action rate is

$$\text{IAR} = \frac{1}{N} \sum_{j=1}^N \mathbb{I}[\pi_{\text{OFF}}^{(E)}(s_j) \in \mathcal{A}_{\text{val}}^+(s_j)], \quad (21)$$

where $\mathcal{A}_{\text{val}}^+(s_j)$ denotes actions consistent with the validated preference at s_j . Together, memory-ON quality, memory-OFF quality, and IAR answer three different questions: whether retrieval helps, whether parameters improve, and whether the learned behavior agrees with verified experience. This audit closes one evolution round and prevents a generic score increase from being misreported as evidence that procedural lessons were internalized.

3.5. Complete evolution round and illustrative trace

The modules above operate in a fixed schedule so that evidence provenance is preserved and the fast and slow paths cannot silently supervise each other. At round e , the accepted multi-role policy $\Pi^{(e)} = \{\pi_r^{(e)}\}_r$ and lesson memory $M^{(e)}$ are held fixed during data collection. The round proceeds as follows:

1. **Collect.** Run participant-disjoint evolve interviews using $(\Pi^{(e)}, M^{(e)})$ and write role-tagged decisions and state transitions to $\mathcal{B}^{(e)}$. No parameter update occurs while trajectories are being collected.
2. **Explore.** Apply Eq. (5) to select hard decisions. For each selected (s, r) , generate K alternatives and replay them with the same frozen profile and simulator configuration.
3. **Verify locally.** Compute the decomposed role utility, invalidate unsafe or inconsistent branches, and apply Eq. (10). Every accepted contrast is written to the role preference buffer $\mathcal{D}_r^{(e)}$ and may independently seed a candidate lesson.
4. **Verify transfer.** Distill candidate lessons, construct source-excluded matched-state sets, and run paired control/intervention replay. Only candidates satisfying Eq. (14) enter $M^{(e+1)}$; rejected candidates and their evidence remain available for audit but never for retrieval.
5. **Consolidate parameters.** Update only roles with at least $N_{\min}$ accepted pairs. Each role update starts from $\pi_r^{(e)}$, uses it as the DPO reference, and freezes the backbone and other adapters.
6. **Accept or reject.** Evaluate the assembled candidate policy with the pre-registered development objective and role-level non-inferiority constraints. If the regression gate fails, set $\Pi^{(e+1)} = \Pi^{(e)}$ while retaining any independently transfer-validated lessons; otherwise accept the candidate adapters. The next collection round begins only after this decision.

This ordering enforces two invariants. First, all action alternatives within a contrast are generated by the same accepted policy round, preventing a later checkpoint from being compared against an earlier policy under different collection conditions. Second, lesson promotion and adapter acceptance are separate transactions. A parameter regression cannot invalidate evidence that a lesson transfers, and a failed lesson abstraction cannot discard a reliable state-local preference.

Consider a sleep-symptom state in which presence is known but frequency remains missing. The interviewer produces four actions from the same state: ask generally about feelings, ask how many nights sleep was difficult during the last two weeks, ask about work impact, or stop the topic. The frozen participant profile gives consistent answers to every branch. The frequency

question fills a scale-critical slot and reduces item-score error at one-turn cost; stopping preserves cost but leaves the item under-determined. If their utility margin passes the local gate, the frequency question and stop action form an interviewer preference pair immediately. The distiller may propose the lesson: *when symptom presence is known but frequency determines severity, ask a short frequency question before a broad impact question, unless a safety concern takes priority*. That statement is still only a hypothesis. It enters lesson memory only if it improves utility across source-excluded states with compatible evidence patterns. Independently, interviewer DPO can learn the local chosen/rejected relation. Subsequent memory-OFF evaluation then tests whether the evolved interviewer asks the validated type of question without receiving the lesson in its prompt.

The trace also illustrates the intended scope of verification. Same-state replay controls the simulator profile and starting state; cross-state replay tests repeatability over other simulated cases. Neither replaces clinical validation, estimates treatment effects, or authorizes autonomous adaptation from real users. The framework’s output is an auditable research policy and a provenance-linked lesson bank whose claims are explicitly limited to the offline assessment environment.

4. Experimental methodology

The empirical design follows a claim–evidence structure: every major mechanism in PsyVEC is associated with a controlled comparison, and all participant-level evaluation data remain isolated from evolution, lesson validation, model selection, and threshold tuning. Because this section specifies experiments before results are available, it defines the protocol and planned analyses without inserting synthetic numerical outcomes.

4.1. Research questions and evaluation protocol

We organize the evaluation around seven research questions (Table 3). RQ1 tests end-to-end effectiveness under a matched interaction budget. RQ2 asks whether the result persists across model families and an external corpus. RQ3 isolates the contribution of contrast generation, verification, and consolidation. RQ4 evaluates whether validation prevents unreliable lessons from entering memory. RQ5 distinguishes retrieval-dependent improvement from parametric internalization and tests role localization. RQ6 measures the quality–computation frontier. RQ7 examines robustness, human-perceived interview quality, and interpretable failure-to-success transitions. The primary endpoint is participant-level PHQ-8 total-score mean absolute error (MAE) under a fixed maximum question budget. All other endpoints are secondary unless explicitly designated otherwise.

4.2. Data partitions and leakage controls

Primary corpus. DAIC-WOZ contains 189 participant interviews and its official partition provides 107 training, 35 development, and 47 test sessions (Gratch et al., 2014). Because labels for the challenge test set are not publicly available, we reserve the official development set as the untouched final test set. The

Table 3: Claim–evidence map for the seven experimental studies. Each headline claim has a direct falsification condition.

Study	Question	Required comparison	Claim weakened if ...
RQ1	Does PsyVEC improve assessment at a matched interview budget?	Full system versus prompt and interactive domain baselines	PHQ MAE does not improve or gains require more questions
RQ2	Does the gain persist across model sizes and data splits?	Qwen3.5-9B versus Qwen3.5-27B; DAIC-WOZ versus E-DAIC	gains occur on only one configuration
RQ3	Which mechanisms are necessary?	Single-factor and staged ablations, including matched-compute random branching	a removed mechanism has no measurable consequence
RQ4	Does verification improve lesson reliability?	Verified versus unvalidated lessons on source-excluded paired states	promotion does not raise lesson precision or reduce harmful transfer
RQ5	Is improvement internalized and role localized?	memory ON/OFF; role-local versus global adapter; DPO versus best-branch SFT	OFF gain is absent or global updating causes no extra regression
RQ6	What does evolution cost?	branch budget, retrieval depth, number of rounds, wall time, tokens, GPU-hours, and peak memory	gains disappear at comparable compute or are Pareto dominated
RQ7	When and why does behavior change?	perturbation tests, blinded human ratings, and paired failure-to-success traces	improvements are simulator-specific, unsafe, or not behaviorally attributable

107 official training participants are stratified by depression status and divided once into an *evolve* set (80%) and a *development* set (20%). The evolve partition is the only partition from which trajectories, hard states, preference pairs, and lesson candidates may be generated. The internal development partition is used for hyperparameter choice, lesson-promotion thresholds, and adapter regression gates. Official-development participants are opened only after the complete protocol is frozen. We additionally repeat the evolve/development split with five preregistered seeds as a split-sensitivity analysis while leaving the final test participants unchanged.

The participant simulator receives a structured profile derived from the source interview, including scale-relevant evidence, permissible paraphrases, and explicit unknown fields. This profile is hidden from every assessment policy. The simulator must answer from the profile rather than invent missing symptoms; branches that contradict the profile or one another are invalidated. Profile creation, lesson validation, and final evaluation are participant-disjoint. Retrieval indices are rebuilt for each split so that no test utterance, profile, embedding, lesson, or cached model response can enter the evolve or development pipeline.

External validation.. E-DAIC is used only as an external evaluation corpus under its authorized release and official participant boundaries. No E-DAIC case is used for evolution or model selection. We report both a direct-transfer setting, in which DAIC-evolved policies and lessons are frozen, and a recalibration setting, in which only scoring thresholds are selected on the permitted E-DAIC development portion. The two settings are reported separately; the recalibrated result is not described as zero-shot transfer.

Simulator audit.. For every branch set, we record profile-consistency rate, cross-branch contradiction rate, refusal rate, and the fraction of branches removed by the validity gate. Common random numbers are used: all methods receive the same participant profile order and simulator-seed schedule. A robustness subset is replayed with a second simulator family so that assessor gains cannot be attributed solely to idiosyncrasies of one patient model. Ground-truth PHQ labels are accessible only

to the offline utility evaluator when score gain is enabled; they are never placed in prompts, state summaries, memories, or retrieval keys.

4.3. Backbones and implementation controls

The planned assessor backbones are Qwen3.5-9B and Qwen3.5-27B, both from the Qwen3.5 series (Qwen Team, 2026). The matched 9B/27B pair reduces model-family confounding and tests whether PsyVEC remains effective at two capacity levels. With only two sizes, however, the comparison is a capacity-sensitivity analysis rather than a scaling law. Both models are evaluated in text-only, non-thinking mode with identical role instructions, evidence schema, context limit, retrieval interface, decoding parameters, and interview-turn budget. A thinking-enabled sensitivity test is reported separately, if used, with reasoning tokens included in the computation total.

All methods use the same frozen participant simulator, case profiles, random-seed manifest, PHQ rubric, and safety policy. Interview actions use deterministic or near-deterministic decoding; exploratory branching alone uses stochastic decoding to produce diverse candidates. PsyVEC uses $K = 4$ branches by default: a low-temperature policy action, an exploratory action, a diversity-prompted action, and, when applicable, a lesson-guided action. Retrieval returns at most three lessons. The maximum number of follow-up questions per PHQ item and all stopping thresholds are selected on the internal development set and then fixed.

The slow path trains one LoRA adapter per role while freezing the backbone and other role adapters. We use the same target modules, rank, update count, effective batch size, and DPO β across backbones whenever memory permits; any quantization or gradient-accumulation difference required by the 27B model is disclosed. A compact development sweep is limited to LoRA rank $\{8, 16\}$, learning rate $\{5 \times 10^{-6}, 10^{-5}\}$, and $\beta \in \{0.05, 0.1\}$. Selection maximizes the development objective in Eq. (19), not final-test MAE. Every experiment is repeated with three end-to-end evolution seeds, and the same accepted checkpoints are reused across all post-hoc analyses.

4.4. Comparison methods and fairness boundaries

The comparison suite contains two prompting controls and four domain-relevant families (Table 4). Each trainable or interactive baseline is rerun on both assessor backbones wherever its mechanism permits, rather than comparing PsyVEC with a published score obtained from a different model, data split, or simulator.

Two fairness boundaries are essential. First, SEGA and SEGA++ analyze completed clinical transcripts and are therefore evaluated in a separate *passive full-transcript* panel. They can contextualize assessment accuracy when all evidence is already available, but they are excluded from question efficiency, evidence acquisition, and self-evolution rankings. Second, MAGI and WiseMind were introduced for protocols broader than PHQ-8. Their PHQ adaptations are labeled explicitly, with every prompt, knowledge-graph mapping, and deviation from the original implementation released. Published native scores are summarized only as background and are never entered into significance tests against PsyVEC.

4.5. Metrics and statistical analysis

Assessment quality is measured by total-score MAE (primary), item-level MAE, quadratic weighted kappa for ordinal scores, and Macro-F1, sensitivity, specificity, and AUROC for the conventional PHQ-8 threshold of 10. Interview quality is measured by critical-evidence coverage, mean valid follow-up questions, tokens per interview, redundant-question rate, premature-stop rate, and the number of acquired critical slots per question. We report an accuracy–cost curve rather than assuming that the system with the longest interview is best.

Evolution quality is measured by locally accepted-pair rate, lesson promotion rate, lesson precision on source-excluded states, negative-transfer rate, memory retrieval hit rate, memory-ON gain, memory-OFF parametric gain from Eq. (20), and internalized-action rate from Eq. (21). Safety metrics include unsafe-question rate, missed escalation-cue rate, contradiction rate, and the worst subgroup change. Computational metrics include assessor and simulator tokens, branch calls, wall-clock latency, GPU-hours, peak GPU memory, trainable parameters, and storage per accepted lesson and preference pair.

All headline comparisons are paired at participant level. We report the mean over three evolution seeds with 95% hierarchical bootstrap confidence intervals, resampling participants at the outer level and run seeds within participant. PHQ MAE differences are additionally tested with a two-sided paired permutation test; depression-label disagreements use McNemar’s test. Ordinal and human-rating analyses use paired bootstrap intervals and an appropriate signed-rank sensitivity test. Holm correction is applied within each family of secondary hypotheses, while the single primary endpoint is left unadjusted. We report absolute differences, relative changes where meaningful, and paired effect sizes rather than significance alone. If the interval crosses zero, the corresponding improvement is described as inconclusive. Missing or invalid simulator branches are reported as outcomes, not silently regenerated until successful.

Table 4: Baseline taxonomy and admissible comparisons. “Adapted” variants preserve the published role logic but map their original diagnostic protocol to the common PHQ-8 evidence schema. They are not presented as reproductions of the papers’ native benchmarks.

Method	Year	Controlled implementation	Information access	Valid comparison
Direct Prompt	–	One interviewer asks and scores from the PHQ rubric; no explicit planning, cross-case memory, or parameter update	Acquired turns only	Accuracy, coverage, turns, safety
Structured Evidence Prompt	–	One agent emits a missing-evidence plan before each action and a structured score rationale; no cross-case learning	Acquired turns only	Effect of stronger prompting
SEGA and SEGA++ (Chen et al., 2024)	2024	Official or faithfully reimplemented structural-element graph and augmentation settings	Complete recorded transcript	Passive assessment reference only
MAGI-PHQ (adapted) (Bi et al., 2025)	2025	Navigation, question, response-judgment, and diagnosis roles; MINI navigation is explicitly mapped to PHQ evidence slots	Acquired turns only	Role decomposition, not native MINI performance
AgentMental (Hu et al., 2026)	2026	Published four-role questioning/evaluation/scoring/updates workflow, rerun with the common backbone and budget	Acquired turns only	Interactive domain baseline
WiseMind-PHQ (adapted) (Wu et al., 2026)	2026	Rational/emotional agents with a PHQ-grounded diagnostic graph and the common stopping budget	Acquired turns only	Knowledge-guided multi-agent behavior
PsyVEC	Ours	Full verified fast and slow consolidation paths	Acquired turns; verified cross-case lessons	Proposed method

5. Experimental studies and analyses

5.1. RQ1: End-to-end assessment effectiveness and efficiency

The main experiment reports methods chronologically: the two non-paper prompting controls, MAGI-PHQ (2025), AgentMental (2026), WiseMind-PHQ (2026), and PsyVEC. All receive the same backbone, participant simulator, maximum PHQ-item follow-ups, and decoding budget, and every system observes only evidence acquired through its own questions. For each backbone, the main table reports participant-level PHQ metrics alongside mean questions, critical-slot coverage, redundancy, and safety. A separate earlier-work panel reports SEGA and SEGA++ (2024) on complete recorded transcripts; it is visually separated and makes no turn-efficiency claim.

The decisive comparison is not unconstrained accuracy but total-score MAE at matched question cost. We therefore evaluate all interactive systems at budgets of one, two, and three follow-ups per item and plot MAE and Macro-F1 against mean questions and generated tokens. PsyVEC supports the main effectiveness claim only if its paired MAE improves without a safety regression and the operating point is not dominated by a baseline that achieves equal or lower error using fewer questions. This experiment also reports per-item results to reveal whether aggregate gain comes only from easily elicited symptoms.

5.2. RQ2: Cross-backbone, cross-corpus, and data-efficiency generalization

We run one full protocol with the 9B backbone (Qwen3.5-9B) and another with the 27B backbone (Qwen3.5-27B). Lessons and LoRA adapters never cross model sizes in the primary setting. The interaction between method and capacity is analyzed using participant-level paired differences: the claim is robustness if the direction of gain is consistent, while any size-related advantage is reported descriptively rather than as a scaling law. A transfer variant retrieves the same validated natural-language lessons across sizes without transferring adapters; this tests whether the fast path is portable within the Qwen3.5 family.

For corpus transfer, DAIC-evolved policies are frozen and evaluated on E-DAIC profiles. We report direct transfer first and threshold recalibration second. To measure label efficiency, evolution is repeated with 25%, 50%, 75%, and 100% of the DAIC evolve participants, using nested subsets shared by all methods.

Curves show performance against the number of evolve participants and the number of accepted contrasts. This analysis tests whether verified consolidation is useful in the small-data regime rather than merely exploiting more synthetic trajectories.

5.3. RQ3: Mechanism ablation under controlled compute

The core ablation is staged so that each row changes one mechanism relative to its parent:

1. interview policy without cross-case consolidation;
2. add unvalidated natural-language lessons;
3. replace them with transfer-validated lessons;
4. add role-local DPO without lesson retrieval;
5. combine verified lessons with one global DPO adapter;
6. full PsyVEC with verified lessons and role-local DPO.

Targeted single-factor tests then address the most contestable design choices. Hard-state mining is compared with random-state branching using exactly the same number of simulator calls. Same-state branching is compared with preference pairs mined from unmatched successful/failed trajectories. The task utility is evaluated both with critical-evidence gain only and with evidence plus hidden score-improvement feedback. Best-branch SFT trains on a^+ alone and is compared with DPO on (a^+, a^-) . Finally, removing the development regression gate quantifies catastrophic or cross-role regressions. For every ablation, we report both test performance and the intermediate quantity the component is designed to change; for example, hard-state mining must improve accepted pairs per branch call, not only terminal MAE.

5.4. RQ4: Contrast validity, lesson precision, and memory governance

This study audits the fast path rather than treating memory as a black box. Candidate lessons are grouped as (i) unvalidated, (ii) promoted by paired cross-state replay, and (iii) rejected or quarantined. On source-excluded matched states, lesson precision is the proportion with positive paired utility, while negative-transfer rate is the proportion with utility below a prespecified harmful margin. We report the full distribution of paired gains, bootstrap intervals, the number of validation states per lesson, and safety outcomes. Verification is supported only if promoted

lessons have higher precision and lower negative transfer than unvalidated lessons.

Threshold sensitivity varies the minimum validation support, win-rate threshold, and promotion margin one at a time. A calibration plot relates estimated validation gain to subsequent held-out gain; severe overconfidence would reveal that the gate is filtering noise without predicting transfer. Governance analysis reports insert, merge, revise, quarantine, and retire operations, memory size by round, duplicate rate, lesson age, retrieval frequency, and the fraction of retrieved lessons that actually change an action. A deliberately useful negative result is retained: locally valid preference pairs whose distilled lessons fail transfer show why the two gates should not be collapsed.

5.5. RQ5: Policy internalization and role-localized credit

Four matched configurations are evaluated for each accepted checkpoint: initial policy with memory OFF or ON, and evolved policy with memory OFF or ON. The same participant states and simulator seeds are reused. The ON difference isolates retrieval benefit, whereas the evolved-versus-initial OFF difference estimates parametric gain. Internalized-action rate is computed only on held-out states satisfying a verified lesson trigger and is accompanied by action-category confusion matrices. We claim policy internalization only when memory-OFF quality and IAR both improve; an ON-only gain is reported as verified memory adaptation.

Role localization is tested by comparing four role-specific adapters with a single global adapter trained on the identical preference pairs and update budget. In addition to total performance, we construct a role-regression matrix: rows identify the updated role and columns report interviewer relevance, stop accuracy, item scoring, and state-revision performance. Selective component replay is evaluated against direct routing alone on ambiguous failures. If a global adapter matches quality without increasing off-target regression, the role-localization contribution is weakened and the simpler alternative is preferred.

5.6. RQ6: Computation–quality and memory–latency trade-offs

Self-evolution introduces costs that endpoint accuracy can conceal. We vary the branch count $K \in \{2, 4, 8\}$, fraction of hard states explored, retrieval depth $k \in \{0, 1, 3, 5\}$, and number of evolution rounds. For each setting, we report accepted contrasts and promoted lessons per 1,000 simulator calls, total generated tokens, GPU-hours, wall-clock time, peak memory, and storage. Random-state and hard-state branching receive identical branch-call budgets. Inference latency is decomposed into role generation, retrieval, and coordination overhead; offline consolidation cost is reported separately so that a fast deployed interviewer is not penalized by hidden training time or vice versa.

The principal visualization is a Pareto frontier of PHQ MAE versus (i) mean interview tokens and (ii) total offline evolution compute. We also report marginal gain per additional 1,000 branch calls and the break-even round after which lesson or adapter growth yields negligible development improvement. This study determines whether the full method is necessary

at all budgets: a memory-only variant may be preferable under tight training resources, while role-DPO may be justified only when repeated deployment amortizes offline cost.

5.7. RQ7: Robustness, human assessment, and failure-to-success transitions

Robustness tests preserve participant ground truth while perturbing surface form: response paraphrase, answer order, hedging, mild contradiction, missing non-critical detail, simulator temperature, and conversation length. A second simulator family is used on a stratified subset. Results are stratified by depression status, PHQ severity band, interview length, and available demographic groups; subgroup intervals and worst-group changes are reported without claiming fairness from underpowered strata. We also test memory poisoning by inserting plausible but rejected lessons and verify that the promotion and retrieval gates prevent their use.

Automatic metrics are complemented by a blinded human study on a stratified sample of paired interviews. At least two raters with mental-health interviewing or clinical annotation training evaluate question relevance, empathy, redundancy, evidence sufficiency, and safety using a fixed rubric. System identity, backbone, and evolution state are hidden; pair order is randomized. We report Krippendorff’s α (or weighted κ for ordinal items), rater-wise scores, adjudication policy, and paired confidence intervals. The study evaluates interaction quality, not clinical diagnostic validity.

Finally, we analyze paired transitions rather than selecting only favorable anecdotes. Every test case is assigned to one of four cells: failure→failure, failure→success, success→failure, or success→success, where success is predefined by item-score correctness and critical-evidence sufficiency at the matched budget. We sample equally from all non-empty cells and trace the hard state, alternative actions, utility components, lesson decision, retrieved memory, and final role output. Aggregating transitions by failure type—missing frequency, redundant question, premature stop, incorrect score, stale update, or simulator inconsistency—reveals which mechanisms repair errors and which create new ones. Success→failure cases receive the same narrative depth as improvements and directly inform lesson retirement and regression-gate design.

5.8. Reproducibility and reporting checklist

The released artifact will include participant identifiers for every split, preprocessing and profile-construction code, prompt templates, PHQ evidence schemas, baseline adaptations, all random seeds, branch caches, accepted and rejected contrasts, lesson provenance, LoRA configurations, checkpoint-selection logs, and scripts that regenerate every table and figure. Hardware model, software versions, precision, quantization, token accounting, GPU-hours, peak memory, and failed runs are reported. All thresholds and primary/secondary endpoints are frozen before final-test evaluation. Because the framework concerns sensitive mental-health interactions, released examples are de-identified and inspected for recoverable personal information; no raw licensed transcript is redistributed.

6. Results and analysis templates

Planning-only material. Every gray table entry and every plotted value in this section is an illustrative placeholder, not an experimental result. Replace the cells and the six `fig_*.csv` files with measured values, revise the surrounding interpretation, and remove this notice before submission.

6.1. *RQ1 template: main comparison and accuracy–cost frontier*

Table 5 is the primary result table. The two prompting controls are listed first because they have no publication year; all research methods then follow publication order. The access column prevents passive full-transcript methods from being mistaken for interactive competitors. Boldface should denote the best interactive result and underline the second-best; significance marks should be added only for preregistered paired comparisons against PsyVEC.

Figure 2 is more informative than a single fixed-budget result: a method is dominated if another achieves lower MAE with no more questions. Replace the two CSV files with the mean at each preregistered turn budget and add bootstrap ribbons when participant-level runs are available.

Table 5: Main assessment results. All displayed values are placeholders. P: passive full-transcript access; I: interactive acquired evidence only. Passive rows are contextual references and are excluded from question-efficiency and paired interactive significance tests. Report mean [95% CI] over three evolution seeds.

Method	Year	Access	Qwen3.5-9B			Qwen3.5-27B		
			MAE ↓	Macro-F1 ↑	Questions ↓	MAE ↓	Macro-F1 ↑	Questions ↓
Direct Prompt	–	I	X.XX	XX.X	XX.X	X.XX	XX.X	XX.X
Structured Evidence Prompt	–	I	X.XX	XX.X	XX.X	X.XX	XX.X	XX.X
SEGA (Chen et al., 2024)	2024	P	X.XX	XX.X	–	X.XX	XX.X	–
SEGA++ (Chen et al., 2024)	2024	P	X.XX	XX.X	–	X.XX	XX.X	–
MAGI-PHQ (Bi et al., 2025)	2025	I	X.XX	XX.X	XX.X	X.XX	XX.X	XX.X
AgentMental (Hu et al., 2026)	2026	I	X.XX	XX.X	XX.X	X.XX	XX.X	XX.X
WiseMind-PHQ (Wu et al., 2026)	2026	I	X.XX	XX.X	XX.X	X.XX	XX.X	XX.X
PsyVEC (ours)	Ours	I	X.XX*	XX.X	XX.X	X.XX*	XX.X	XX.X

*Paired permutation $p < 0.05$ against the strongest interactive baseline after Holm correction. Replace this note if the actual test is not significant. Add item MAE, QWK, AUROC, evidence coverage, redundancy, and safety in an appendix table rather than widening the headline table.

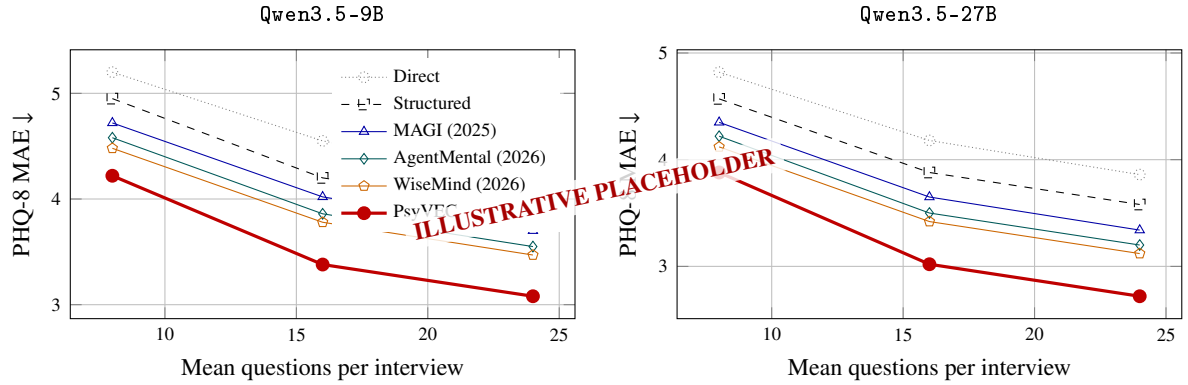


Figure 2: Accuracy–interaction–cost frontier. The plotted values are illustrative placeholders loaded from `fig_accuracy_cost_9b.csv` and `fig_accuracy_cost_27b.csv`. Replace them with observed means and 95% confidence bands.

6.2. RQ2 template: capacity, corpus transfer, and data efficiency

For the data-efficiency plot, use evolve-set fractions 25%, 50%, 75%, and 100% on the horizontal axis and show both test MAE and accepted contrasts. Nested subsets and identical participant IDs across methods are required; otherwise, apparent sample-efficiency differences are confounded by case composition.

Table 6: Capacity and corpus-transfer decomposition. Placeholder cells should contain paired improvement of PsyVEC over the strongest valid interactive baseline, not raw cross-protocol rankings.

Training/evolution source	Evaluation condition	MAE ↓	ΔMAE	Macro-F1 ↑	Questions	Safety regressions	Interpretation
DAIC, Qwen3.5-9B	DAIC held-out	X.XX	−X.XX	XX.X	XX.X	0/X	In-domain small model
DAIC, Qwen3.5-27B	DAIC held-out	X.XX	−X.XX	XX.X	XX.X	0/X	In-domain large model
DAIC, Qwen3.5-9B	E-DAIC direct transfer	X.XX	−X.XX	XX.X	XX.X	0/X	No target calibration
DAIC, Qwen3.5-27B	E-DAIC direct transfer	X.XX	−X.XX	XX.X	XX.X	0/X	No target calibration
DAIC, Qwen3.5-27B	E-DAIC recalibrated	X.XX	−X.XX	XX.X	XX.X	0/X	Label explicitly as recalibrated

6.3. RQ3 template: mechanism and matched-compute ablations

Table 7: Staged and targeted ablations. Gray entries are placeholders. Each row must reuse the same split, seed schedule, and branch-call budget as its parent unless the changed factor is explicitly “more compute.”

Variant	Lesson	Transfer gate	DPO	Role-local	Hard-state	MAE ↓	Pair yield ↑	Cross-role regress. ↓
No cross-case evolution	–	–	–	–	–	X.XX	–	–
+ Unvalidated lessons	✓	–	–	–	–	X.XX	–	X.X%
+ Verified lessons	✓	✓	–	–	–	X.XX	–	X.X%
Role-DPO only	–	–	✓	✓	✓	X.XX	XX.X%	X.X%
Verified lesson + global DPO	✓	✓	✓	–	✓	X.XX	XX.X%	X.X%
Full PsyVEC	✓	✓	✓	✓	✓	X.XX	XX.X%	X.X%
Random-state branching (matched calls)	✓	✓	✓	✓	–	X.XX	XX.X%	X.X%
Unmatched trajectory preferences	✓	✓	✓	✓	✓	X.XX	XX.X%	X.X%
Evidence-only utility	✓	✓	✓	✓	✓	X.XX	XX.X%	X.X%
Best-branch SFT	✓	✓	SFT	✓	✓	X.XX	XX.X%	X.X%

6.4. RQ4 template: verification and memory quality

The associated calibration plot should place estimated validation gain on the horizontal axis and held-out paired gain on the vertical axis, with one point per lesson and a reliability curve. A high promotion precision with poor calibration means that the gate filters harmful lessons but should not be interpreted as estimating effect magnitude.

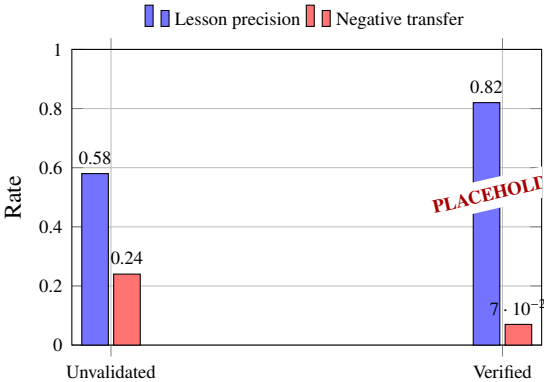


Figure 3: Verification audit template. Illustrative values are loaded from `fig_verification.csv`. Report participant- and lesson-level bootstrap intervals, support per lesson, and the rejected/quarantined distribution.

6.5. RQ5 template: memory internalization and role localization

Table 8: Memory-ON/OFF decomposition and role-localization audit. The four policy–memory configurations must use identical held-out states and simulator seeds.

Policy	Memory	MAE ↓	Macro-F1 ↑	IAR ↑	Retrieval change	Interviewer	Evaluator	Scorer/Updater
Initial	OFF	X.XX	XX.X	XX.X%	–	baseline	baseline	baseline
Initial	ON	X.XX	XX.X	XX.X%	Δ X.XX	Δ X.X	Δ X.X	Δ X.X
Evolved, global adapter	OFF	X.XX	XX.X	XX.X%	–	Δ X.X	Δ X.X	Δ X.X
Evolved, role-local	OFF	X.XX	XX.X	XX.X%	–	Δ X.X	Δ X.X	Δ X.X
Evolved, role-local	ON	X.XX	XX.X	XX.X%	Δ X.XX	Δ X.X	Δ X.X	Δ X.X

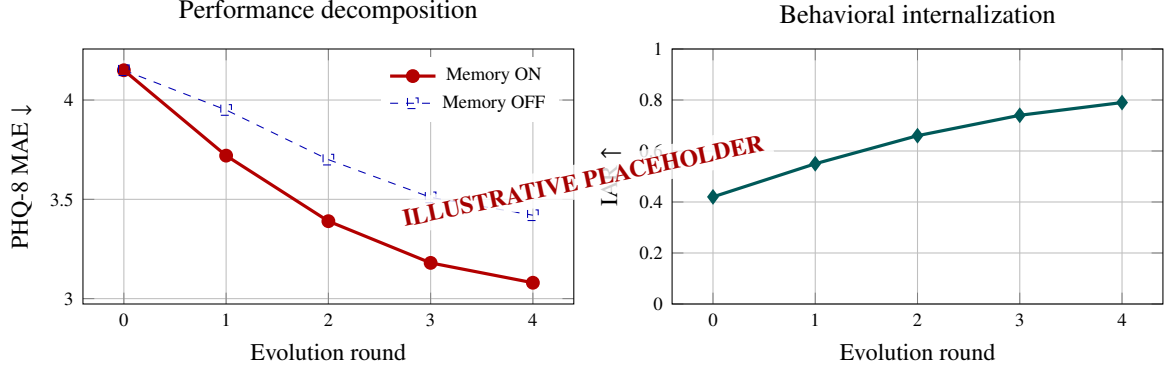


Figure 4: Evolution-round template loaded from `fig_evolution_rounds.csv`. Memory-OFF MAE and IAR must both improve before claiming parametric internalization.

6.6. RQ6 template: computation–quality frontier

The recommended compute table reports, for every variant, branch calls, accepted pairs per 1,000 calls, promoted lessons, trainable parameters, GPU-hours, peak memory, inference latency, and MAE. Report offline training and online interview cost separately, and identify Pareto-dominated settings rather than selecting $K = 4$ after viewing only accuracy.

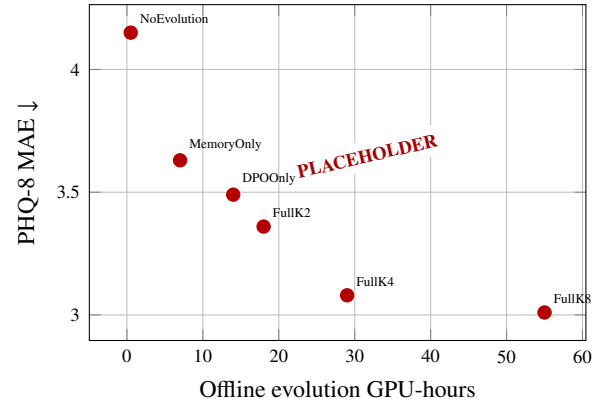


Figure 5: Compute–quality template loaded from `fig_compute_tradeoff.csv`. Add total simulator/assessor tokens, wall time, peak GPU memory, and a second frontier against online interview tokens.

6.7. RQ7 template: human ratings and failure transitions

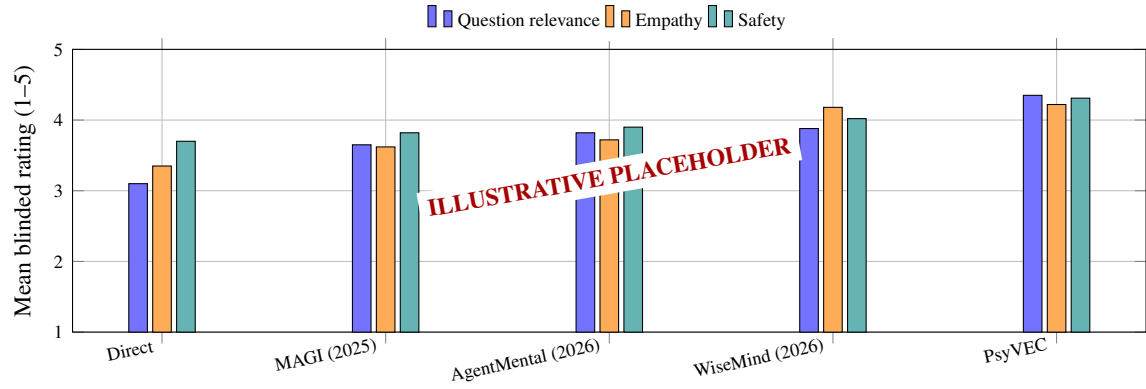


Figure 6: Blinded human-rating template loaded from `fig_human_ratings.csv`. Add 95% paired confidence intervals and report Krippendorff’s α or weighted κ in the caption or accompanying table.

Table 9: Failure-transition matrix at the matched question budget. Report counts with row percentages; never show only failure→success examples.

	Evolved failure	Evolved success	Total
Initial failure	XX (XX.X%)	XX (XX.X%)	XX
Initial success	XX (XX.X%)	XX (XX.X%)	XX
Total	XX	XX	N

Table 10: Paired qualitative trace template. Sample cases from every non-empty transition cell with a fixed rule, then link the behavioral change to a verified contrast or explicitly mark it as unattributed.

Case / transition	Failure type	Missing / known evidence	Initial action	Retrieved lesson or policy change	Evolved action and outcome
ID; F→S	Missing frequency	Presence known; frequency missing	Broad impact question	Frequency-before-impact lesson; interviewer adapter	Targeted frequency question; correct item score
ID; S→F	Premature stop	Ambiguous persistence	Useful follow-up	Overgeneralized stop lesson	Stopped early; lesson quarantined
ID; F→F	Simulator conflict	Contradictory branch answers	Invalid comparison	No accepted contrast	Failure retained; no unsupported claim

The final Results narrative should follow the same order as the seven research questions. For each subsection: state the finding with uncertainty, point to the corresponding table or figure, interpret the mechanism-level metric, report the relevant negative result, and end with the exact scope of the supported claim. This prevents a long collection of metrics from obscuring which contribution each experiment actually tests.

7. Conclusion

To be completed after the empirical findings are available.

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